



IoT-BASED SMART IRRIGATION SYSTEM WITH SOLAR ENERGY INTEGRATION AND PROGRAMMABLE TIMER

BACHELOR IN INFORMATION TECHNOLOGY (HONOURS)(INTERNET OF THINGS)

FINAL YEAR PROJECT



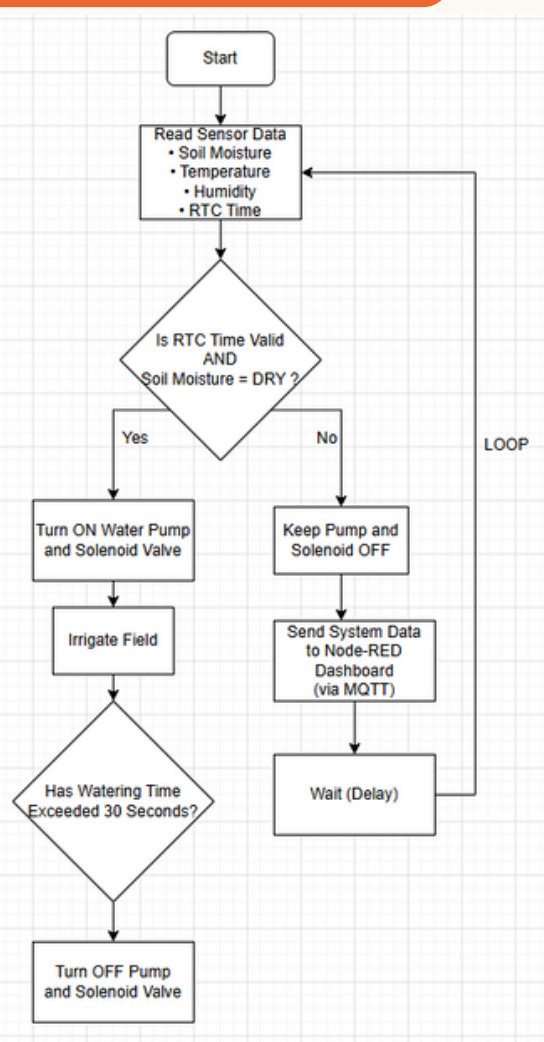
INTRODUCTION

THIS PROJECT DEVELOPS A LOW-VOLTAGE, SOLAR-POWERED SMART IRRIGATION SYSTEM USING IOT TECHNOLOGY TO MONITOR SOIL MOISTURE AND ENVIRONMENTAL CONDITIONS IN REAL TIME. IRRIGATION IS AUTOMATICALLY ACTIVATED ONLY WHEN THE SOIL BECOMES DRY, MAKING IT SUITABLE FOR PLANTS WITH LOW WATER REQUIREMENTS. THE SYSTEM IS DESIGNED FOR SMALL-SCALE AND OFF-GRID APPLICATIONS, PROVIDING AN EFFICIENT AND AUTOMATED SOLUTION FOR BASIC IRRIGATION CONTROL.

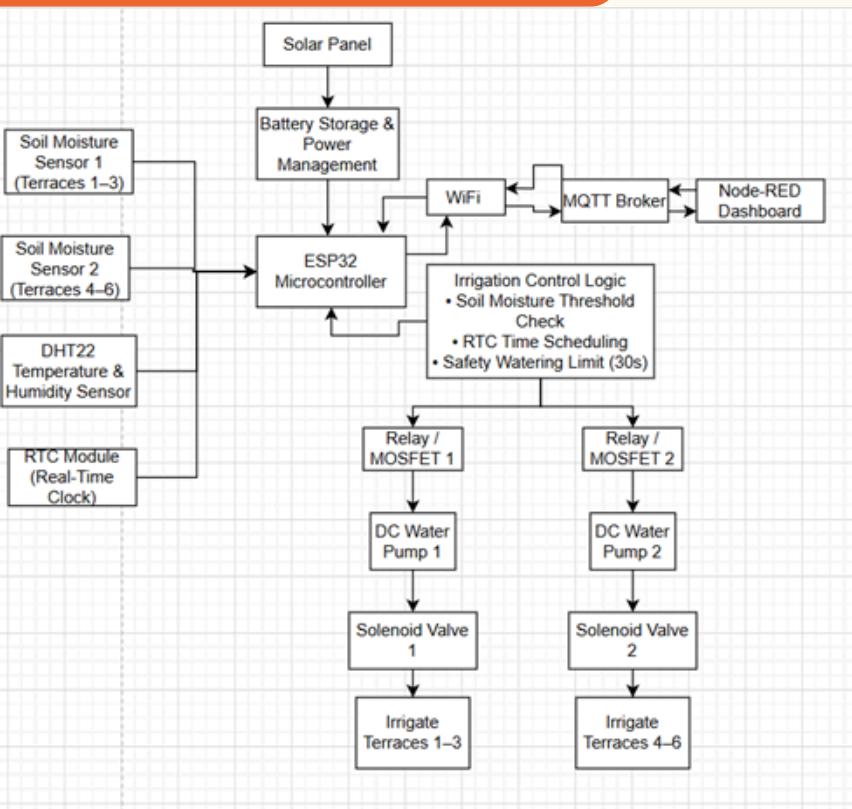
OBJECTIVE

- To develop a low-voltage, solar-powered IoT-based irrigation system suitable for off-grid farming applications.
- To monitor soil moisture, temperature, and humidity in real time using sensors.
- To implement automatic irrigation control based on soil moisture levels with time-based scheduling using an RTC module.
- To enable remote monitoring and manual control through a Node-RED dashboard.

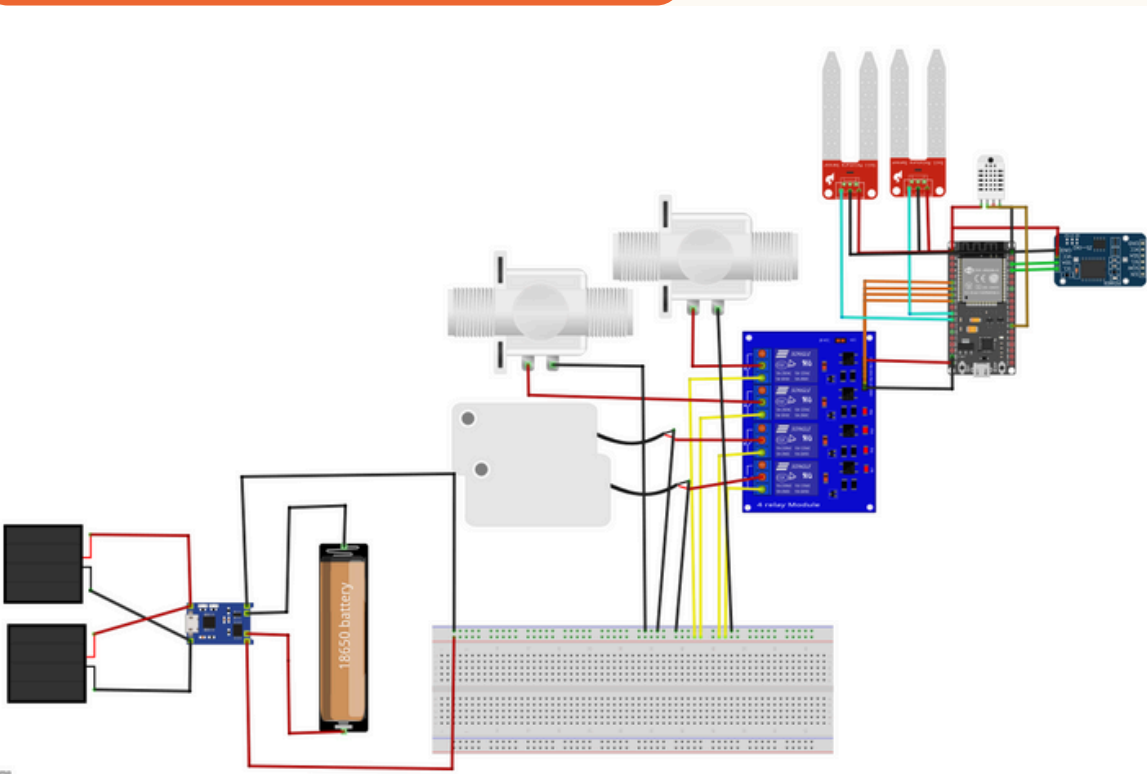
FLOWCHART



BLOCK DIAGRAM



CIRCUIT DIAGRAM



CONCLUSION

This project demonstrates a low-cost, solar-powered smart irrigation system that uses IoT technology and real-time sensor data to automate irrigation for small-to-medium farms. The system integrates soil moisture sensing, environmental monitoring, and scheduled control to support efficient water usage while reducing manual labor. By utilizing renewable energy and web-based monitoring through Node-RED, the solution is suitable for off-grid agricultural environments. Although challenges such as connectivity limitations and prototype scalability exist, the project highlights the potential of affordable IoT-based irrigation systems to support smarter and more sustainable farming practices.

PROBLEM STATEMENT

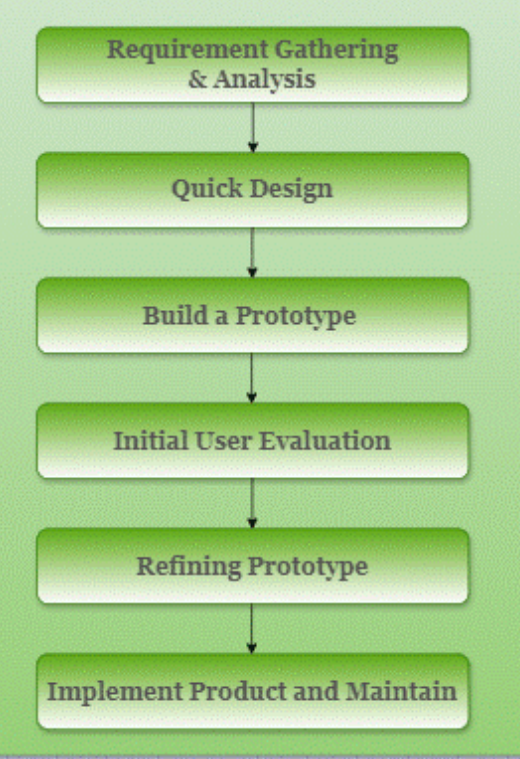
MANY SMALL-TO-MEDIUM FARMS STILL DEPEND ON MANUAL OR FIXED-TIME IRRIGATION SYSTEMS, WHICH CAN RESULT IN INEFFICIENT WATER USAGE, INCONSISTENT CROP GROWTH, INCREASED LABOR EFFORT, AND POWER LIMITATIONS IN REMOTE AREAS. THE HIGH COST AND COMPLEXITY OF EXISTING SMART IRRIGATION SOLUTIONS FURTHER RESTRICT THEIR ADOPTION. THEREFORE, THERE IS A NEED FOR A LOW-COST, LOW-VOLTAGE, SOLAR-POWERED IOT-BASED IRRIGATION SYSTEM THAT CAN AUTOMATICALLY RESPOND TO SOIL CONDITIONS AND SUPPORT REMOTE MONITORING.

SCOPE & LIMITATIONS

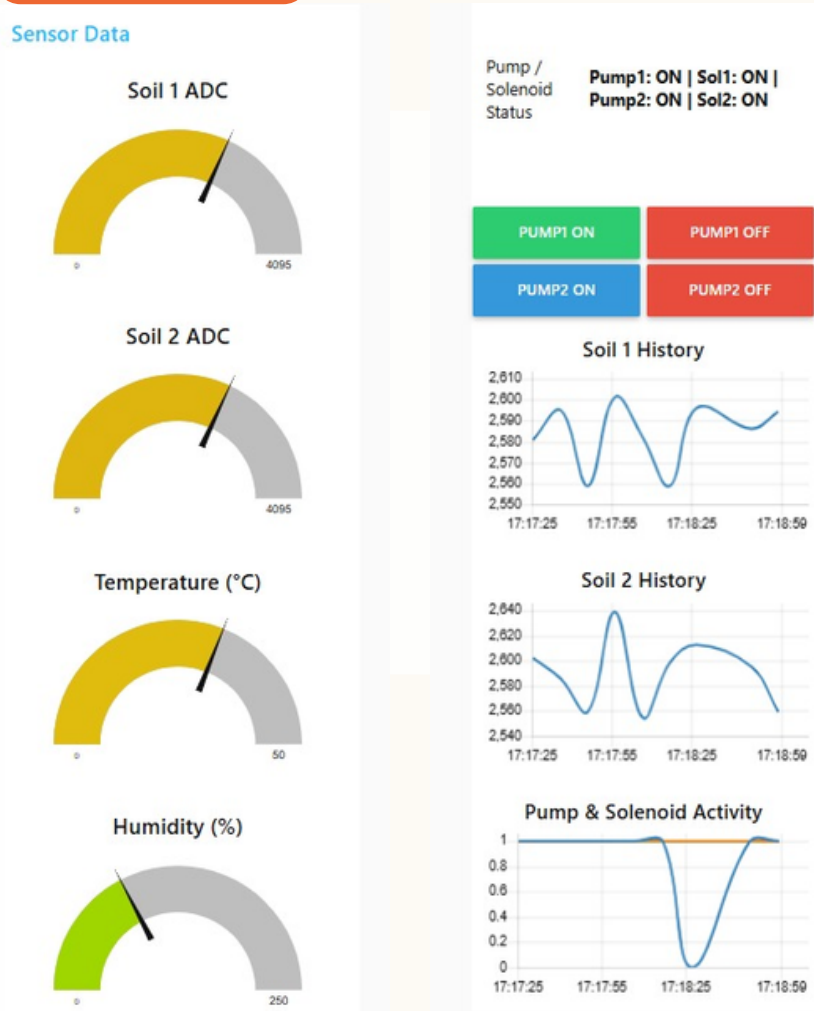
THIS PROJECT DEVELOPS A SOLAR-POWERED, IOT-BASED SMART IRRIGATION PROTOTYPE FOR SMALL-TO-MEDIUM OFF-GRID FARMS, FEATURING REAL-TIME MONITORING, AUTOMATIC IRRIGATION BASED ON SOIL MOISTURE, AND REMOTE CONTROL VIA NODE-RED DASHBOARD. THE SYSTEM IS LIMITED BY ENVIRONMENTAL FACTORS, CONNECTIVITY ISSUES, INITIAL HARDWARE COST, AND SCALABILITY BEYOND THE PROTOTYPE SETUP.

METHODOLOGY

Prototyping Model



RESULT



LITERATURE REVIEW

Author(s)	Year	System Description	Technologies Used	Unique Features
Huque et al.	2023	IoT-based system using NodeMCU for real-time monitoring	NodeMCU, soil moisture, temperature, and humidity sensors	Automates irrigation, optimizes water use
Rehman et al.	2023	IoT system for soil wetness detection with mobile monitoring	Microcontrollers, cloud services, mobile app	Remote access and automatic watering adjustment
Falchetta et al.	2023	Solar-powered irrigation for small-scale farms in Africa	Solar PV systems	Reduces fossil fuel reliance, supports remote areas
John Hutton	2025	Market review on solar-powered irrigation systems	Solar technology	Focuses on global trends and government support
Bahubalendruni et al.	2024	Smart irrigation with sensor-actuator integration	IoT sensors, actuators, remote control interface	Automates schedule, user-friendly dashboard
Agyeman et al.	2023	Predictive irrigation scheduler using AI	ML algorithms (k-means, LSTM), sensors	Forecasts moisture levels, improves yield

REFERENCES

Huque, M. T., Godhuli, J. J., Raziur, S. M., Pushon, R., & Haque, E. (2023). Internet of Things (IoT)-based Smart Irrigation System for Sustainable Agriculture. *Journal of Informatics Electrical and Electronics Engineering (JIEEE)*, 4(3), 1-7.
<https://doi.org/10.54060/JIEEE.2023.96>